

SIMATS ENGINEERING

Department of Computer Science and Engineering



CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA1512
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	T.PRAKEERTHI
Register Number	192524279
Class / Slot	Slot B
Capstone Title	Medi Queue AI: Hospital of Queue and No-show. prediction App.
Capstone Product Type	Web app / Android app / iOS app / Hybrid app
CO1 Architecture Reference	Mention concept map / presentation title / GitHub link if available

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must Medi Queue AI: Hospital of Queue and No-show. prediction App.support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	Patients, Doctors, Hospital Receptionists, Hospital Admin
User roles	Patient – books appointments; Doctor – views schedule; Receptionist – manages queue; Admin – monitors reports
Main modules	Patient Registration, Appointment Booking, Queue Management, No-show Prediction, Notifications, Dashboard
Cloud services identified in CO1	Cloud Storage, Cloud Database, Virtual Machine (VM), API Service, AI Analytics
API/backend need	REST API for appointment booking, queue updates, notifications, and AI prediction
Database need	Stores patient details, appointments, queue status, doctor schedules, and prediction results
File upload/storage need	Store medical reports, prescriptions, and patient documents securely
Analytics/AI/Python module	AI/Python model predicts patient no-shows and estimates waiting time using appointment history

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: My capstone workload type and reason

Workload Type: AI/Python Module + API Backend + Database-heavy App

Reason:

The app uses AI to predict patient no-shows, APIs for appointment and queue management, and a database to store patient, doctor, and appointment records. Therefore, it is mainly an AI/Python-based analytics application with API and database support.

Starting VM Guidance:

4–8 vCPU, 16–32 GB RAM (GPU optional)

This is the appropriate answer for your Medi Queue AI: Hospital Queue and No-show Prediction App.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Current Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000

Input	Value	Reason / Assumption
Registered users expected	1000	Estimated hospital patients
Active user percentage	10%	Around 10% active during peak hours
Peak factor	2	Morning and evening rush
Actions per user per minute	2	Book appointment and check queue
Calculated RPS	6.67RPS	$(1000 \times 10\% \times 2 \times 2) \div 60 = 6.67$
Concurrent active users	200	$1000 \times 10\% \times 2 = 200$
Daily requests	6000	$200 \times 30 \text{ actions per day} = 6000$

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver		Score	
Basic web/API application		1	
Authentication and role management		+1	
Moderate API traffic above 5 RPS		+1	
Database-heavy search or reporting		+1 to +2	
Background jobs or notifications		+1	
Analytics or dashboards		+2	
Python inference inside server		+2 to +4	
Total CPU Score		Recommended vCPU	
1-2		1-2 vCPU	
3-4		2 vCPU	
5-6		4 vCPU	
7-9		4-8 vCPU or separate analytics/AI service	
10+		Scale out with multiple VMs or managed compute	
CPU Driver Selected		Score	Reason
Basic web/API application		2	Main application handles patient requests.
Authentication and roles		1	Patient, doctor, and admin login.
API traffic		2	Moderate number of users during peak hours.
Database/search/reporting		2	Stores appointments and generates reports.
Background jobs		1	Sends reminders and predicts no-shows.
Analytics/dashboard		1	Displays queue status and statistics.
Python module		1	AI model for no-show prediction.
Total CPU Score		10	Sum of all scores.
Recommended vCPU		4vCPU	Suitable for a medium-sized AI web application.

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula

Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2GB
Web/API service	1-2 GB	2GB
Small database	2-4 GB	4GB
Medium database/cache	4-8 GB	4GB
Analytics/reporting	4-8 GB extra	4GB
AI/Python module	4-16 GB extra	4GB
25% buffer	Subtotal x 0.25	5GB
Final RAM	Round up to standard size	24GB

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
OS and application	20-40 GB	30GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	10GB
File uploads	Avg file MB x files/day x days x 1.3	10GB
Logs	Requests/day x log KB x days / 1024 MB	5GB
Backup/growth buffer	Subtotal x 30%	17GB
Final storage	Round up to standard disk size	72GB

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	NO
Two-tier design	App VM + managed/separate database	YES
Three-tier design	Frontend, backend/API and database separated	NO
Containerized backend	Microservices, portable API, CI/CD	NO
Serverless function	Event-based task, notification, small automation	NO
Managed database/storage	Reliability, backup and scaling required	NO

Final VM Plan: selected plan, vCPU, RAM, storage and reason

Selected Plan: Two-tier Design

vCPU: 4 vCPU

RAM: 24 GB

Storage: 80 GB

Reason: Two-tier architecture is suitable for the Medi Queue AI application because it separates the application server and database, providing better performance, security, and scalability for hospital queue management and AI-based no-show prediction.

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments
Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment

Context	Selected Approach	Justification
Local prototype/testing	Type 2 Hypervisor	Easy for local development and testing.
Cloud pilot deployment	Cloud-managed virtualization	Simple cloud deployment and management.
Production growth scenario	Type 1 Hypervisor	Better performance and scalability.
Module suitable for container	Container runtime	Ideal for the AI prediction API and backend services.

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Before every deployment	Quick rollback if needed.
Before database schema change	Snapshot	Before every database schema change	Restore database if schema update fails.

Before OS/package update	Snapshot	Before every OS/package update	Recover from failed OS/package updates.
Before review/demo	Snapshot	Before every review/demo	Restore demo environment if needed.
Create testing environment	Snapshot	Before testing	Safe testing without affecting the main VM.
Create team demo copy	clone	When a demo copy is required	Independent copy for team demonstration.

Technical Tip

A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning
Capstone Module	VM / Container / Managed Service	Reason
Frontend	Container	Easy deployment and scaling.
Backend/API	Container	Fast and portable microservices.
Database	Managed Service	Reliable backup and high availability.
File storage	Managed Service	Secure and scalable storage.
AI/Python module	Container	Easy to run AI prediction models.
Analytics/dashboard	Container	Simple deployment and scaling.
Notification/background jobs	Container	Efficient background processing.

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

Use a **medium-size VM** for development and testing.

Choose a **Type 2 hypervisor** for local development and a **Type 1 hypervisor** for production.

Deploy the frontend, backend, AI module, analytics, and notification services using **containers**.

Use a **managed database service** for secure and reliable data storage.
Store reports and files in **managed cloud storage**.
Create **snapshots** before deployments, database changes, and OS updates.
Create **clones** for testing and team demonstrations.
This architecture provides good performance, scalability, security, and easy maintenance for the Medi Queue AI application.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.